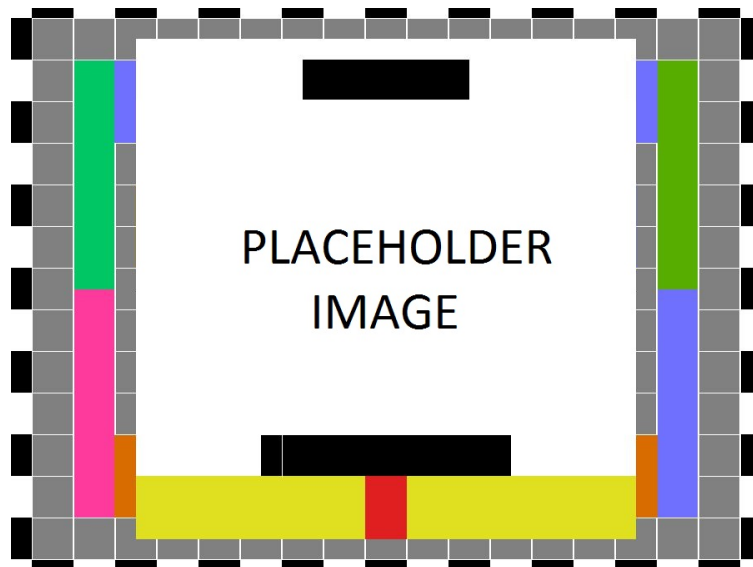


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**ARCHITECTURAL DESIGN SPECIFICATION
CSE 4316: SENIOR DESIGN I
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**TEAM TRAILBLAZER
TERRASCOUT AUTONOMOUS GROUND ROVER**

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REVISION HISTORY

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1 INTRODUCTION

This ADS describes the architecture of TerraScout Autonomous Ground Rover as an architecture description per ISO/IEC/IEEE 42010:2022 [1], using design viewpoints from IEEE Std 1016-2009 [2]. It refines the requirements baselined in the SRS and the concept in the Project Charter. Scope: the rover, its ground-station link, and its charging dock.

2 STAKEHOLDERS & ARCHITECTURAL CONCERNS

Stakeholder	Primary concerns
Sponsor (O&M)	inspection coverage, fault-detection quality, cost, safety
Field technician	simple operation, reliability, safe behavior near people
Development team	feasibility, modularity, testability, integration risk
Maintainers	diagnosability, replaceable detection model, documentation
Instructor/safety	compliance with lab and robot-safety policy

3 ARCHITECTURE VIEWPOINTS

Viewpoint	Frames	Realized in
Context	boundary, external actors	Section 4
Composition/Data Flow	layer/subsystem decomposition, data	Section 5
Logical/Functional	subsystem responsibilities & interfaces	Sections 6–8
Deployment	software-to-hardware mapping	Section 8

3.1 ARCHITECTURAL STYLE & CONVENTIONS

The system uses a three-layer robotics stack: **X — Perception & Sensing**, **Y — Autonomy & Compute**, and **Z — Mobility & Power**. Subsystems communicate over a ROS 2 message bus on the compute module; hardware interfaces use USB, CSI, and CAN. Interface identifiers use #NN. Layers are named by function; subsystems are named [Layer] . [Name].

4 SYSTEM OVERVIEW & CONTEXT VIEW

External actors: the operator (via the ground-station web page and WebSocket link), the physical PV field (sensed environment), and the charging dock (docking + power). The X layer converts the physical world into data; the Y layer turns data into decisions and reports; the Z layer turns decisions into motion and manages power.

5 SUBSYSTEM DEFINITIONS & DATA FLOW

Sensor streams (X) feed localization and perception; the autonomy layer (Y) fuses pose, plans paths, detects faults, and drives the mission state machine, emitting velocity commands to the platform (Z) and status/reports to the ground station.

6 X LAYER SUBSYSTEMS — PERCEPTION & SENSING

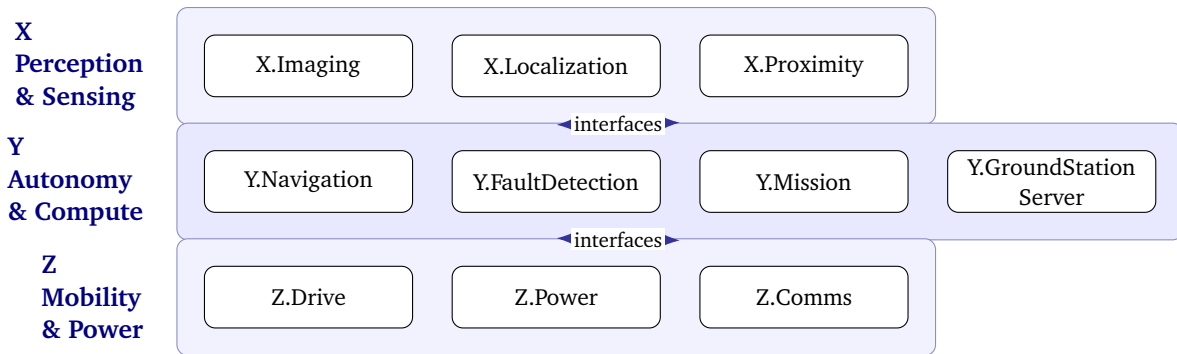


Figure 1: Three-layer architecture of the rover, showing each layer's subsystems

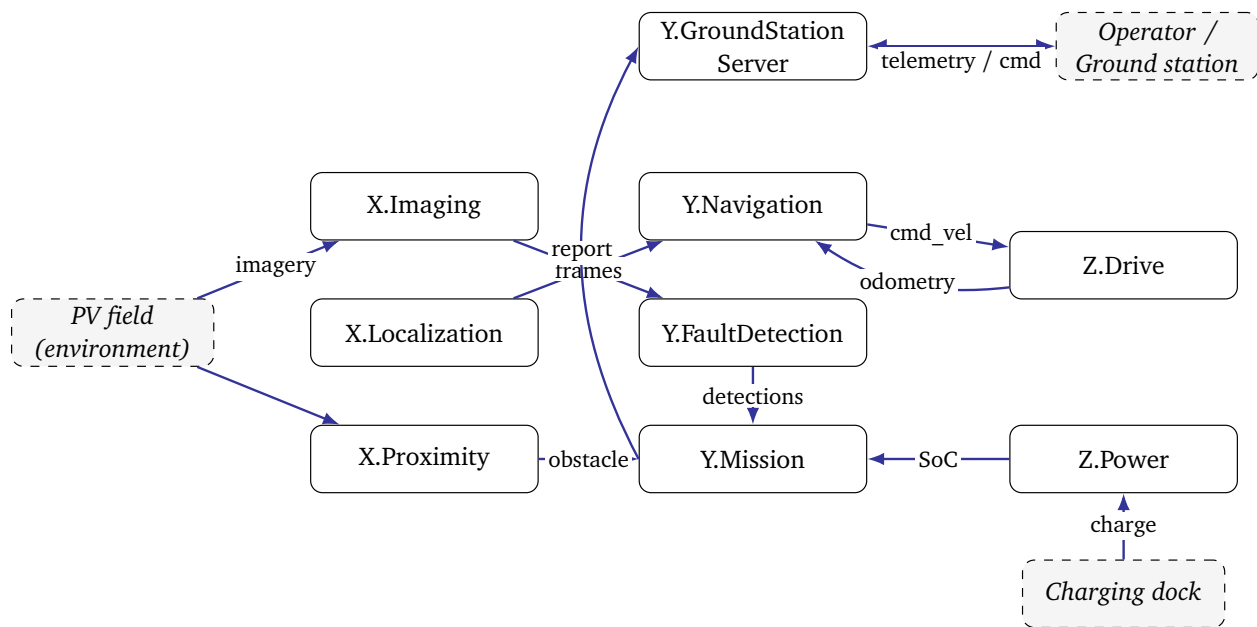


Figure 2: Subsystem data flow: sensing (X) → autonomy (Y) → mobility (Z), with the ground station and dock as external actors

6.1 X.IMAGING

Responsibilities: capture synchronized RGB and thermal frames and tag them with time and pose.

Assumptions: thermal camera provides a USB/CSI stream.

ID	Description	Inputs	Outputs
#01	Camera capture bus	trigger, pose, stamp	RGB frame, thermal frame

Design rationale: on-camera hardware sync avoids frame-to-pose skew (AD-02).

6.2 X.LOCALIZATION

Responsibilities: produce a fused pose from RTK-GNSS and IMU at ≥ 20 Hz.

ID	Description	Inputs	Outputs
#02	Pose estimate	GNSS fix, IMU	fused pose

6.3 X.PROXIMITY

Responsibilities: detect near obstacles (ultrasonic/bumper) and raise a stop signal. Feeds FR-05 safe-stop.

7 Y LAYER SUBSYSTEMS — AUTONOMY & COMPUTE

7.1 Y.NAVIGATION

Responsibilities: plan and follow paths through the waypoint list within the FR-01 tolerance, avoiding obstacles; emit velocity commands.

ID	Description	Inputs	Outputs
#03	Motion command	pose, waypoints, proximity	velocity cmd

7.2 Y.FAULTDETECTION

Responsibilities: classify panel regions in captured frames as nominal or suspected fault (FR-03) and record locations. Model is replaceable (QA-01).

7.3 Y.MISSION

Responsibilities: run the mission state machine (idle→patrol →capture→return→dock), enforce FR-05 triggers, and assemble the FR-04 report.

7.4 Y.GROUNDSTATIONSERVER

Responsibilities: terminate the TLS WebSocket (EI-01, SE-01), stream telemetry, accept commands, and serve reports to the operator page.

8 Z LAYER SUBSYSTEMS — MOBILITY & POWER

8.1 Z.DRIVE

Responsibilities: execute velocity commands via brushless motor controllers over CAN; enforce E-stop (SA-01) in hardware.

ID	Description	Inputs	Outputs
#04	Drive bus (CAN)	velocity cmd	wheel torque, odometry

8.2 Z.POWER

Responsibilities: manage the battery, report state-of-charge (drives FR-05 low-battery), and control dock charging (EI-02).

8.3 Z.COMMS

Responsibilities: provide the Wi-Fi link between the compute module and the ground station. **Deployment:** compute module (Y, X software) mounts on the chassis; Z subsystems are embedded microcontrollers on the CAN bus.

9 ARCHITECTURE DECISIONS & RATIONALE

9.1 AD-01: ROS 2 MESSAGE BUS

Context: multiple concurrent sensing/decision processes. **Decision:** adopt ROS 2. **Alternatives:** custom IPC (more work), ROS 1 (EOL). **Consequences:** mature tooling and modularity (supports QA-01) at the cost of a runtime dependency. **Affects:** FR-01, FR-03, QA-01.

9.2 AD-02: ON-BOARD (EDGE) FAULT DETECTION

Context: FR-03 plus SE-02 (imagery stays on the private network). **Decision:** run detection on-board rather than streaming imagery off-site. **Alternatives:** cloud inference (violates SE-02, needs egress). **Consequences:** bounded by compute budget; model kept replaceable. **Affects:** FR-03, SE-02, PE-03.

10 REQUIREMENTS TRACEABILITY

Req ID	Requirement (abbrev.)	Realizing subsystem(s)
FR-01	Waypoint patrol	Y.Navigation, X.Localization
FR-02	Imagery capture	X.Imaging
FR-03	Fault detection	Y.FaultDetection
FR-04	Inspection report	Y.Mission, Y.GroundStationServer
FR-05	Safe-stop & return	Y.Mission, X.Proximity, Z.Power
SE-01	Encrypted link	Y.GroundStationServer
SA-01	Emergency stop	Z.Drive

REFERENCES

- [1] ISO/IEC/IEEE, “ISO/IEC/IEEE 42010:2022 — Software, systems and enterprise — Architecture description,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2022.
- [2] IEEE, “IEEE Std 1016-2009 — IEEE Standard for Information Technology — Systems Design — Software Design Descriptions,” Institute of Electrical and Electronics Engineers, New York, NY, USA, Tech. Rep., 2009.